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Assignment #1

Algorithm: `generate_subsets(set, current_subset, index)`

```
1 IF (index == LENGTH(set)):
2     PRINT current_subset
3     RETURN
4 generate_subsets(set, current_subset, index + 1)
5 generate_subsets(set, current_subset + set[index], index + 1)
```

When the function undergoes recursion, the runtimes are as follows:

```
1 IF (index == LENGTH(set)): 1
2     PRINT current_subset 0
3     RETURN 0
4 generate_subsets(set, current_subset, index + 1) T(n-1)
5 generate_subsets(set, current_subset + set[index], index + 1) T(n-1)
```

$$\therefore T(n) = 1 + 2T(n-1).$$

When the function terminates, we arrive at $T(0)$ because there are no more elements to process.

The runtimes are as follows:

```
1 IF (index == LENGTH(set)): 1
2     PRINT current_subset n (worst case)
3     RETURN 1
4 generate_subsets(set, current_subset, index + 1) 0
5 generate_subsets(set, current_subset + set[index], index + 1) 0
```

$$\therefore T(0) = 2 + n.$$

Expanding out the terms, we get:

$$\begin{aligned} T(n) &= 1 + 2T(n-1) \\ &= 1 + 2[1 + 2T(n-2)] \\ &= 1 + 2 + 2^2T(n-2) \\ &= 1 + 2 + 2^2[1 + 2T(n-3)] \\ &= 1 + 2 + 2^2 + 2^3T(n-3) \\ &\vdots \\ &= (2^0 + 2^1 + 2^2 + \dots + 2^{k-1}) + 2^kT(n-k) \\ &= \sum_{m=0}^{k-1} 2^m + 2^kT(n-k) \\ &= (2^k - 1) + 2^kT(n-k). \end{aligned}$$

$$T(0) \rightarrow n - k = 0;$$

$$n = k.$$

$$\begin{aligned} \therefore T(n) &= 2^n - 1 + 2^nT(0) \\ &= 2^n - 1 + 2^n(2 + n) \\ &= 2^n(3 + n) - 1. \end{aligned}$$